Relational Databases

You are a data engineer/analyst for an online marketplace.

The platform has customers, sellers, products, orders, and order line-items.

Tables

customers

column	type
customer_id	INT
full_name	VARCHAR
city	VARCHAR
signup_date	DATE

sellers

column	type
seller_id	INT
seller_name	VARCHAR
tier	VARCHAR

orders

column	type
order_id	INT
customer_id	INT
order_date	DATE
status	VARCHAR

order_items

column	type
order_id	INT
product_id	INT
quantity	INT
unit_price	DECIMAL(10,2)
discount_pct	DECIMAL(5,2)

products

column	type
product_id	INT
seller_id	INT
category	VARCHAR
list_price	DECIMAL

Metrics

1. Line revenue = quantity * unit_price * (1 - discount_pct/100)
2. Order revenue is the sum of its line items.

Questions

1. Return all completed orders with columns: order_id, customer_id, order_date. Sort by order_date ascending.
2. List all orders with the customer name. Include cancelled/returned too.
3. Compute total number of completed orders per customer. Customers with 0 completed orders must appear with 0.
4. Compute revenue per completed order.
5. Compute total revenue per seller (completed orders only).
6. For each customer, number their completed orders by order_date ascending. Display customer_id, order_id, order_date, cust_order_seq.
7. Rank sellers by total revenue on completed orders (same revenue definition as earlier). Display seller_id, seller_name, revenue, revenue_rank.
8. For each customer's completed orders (by order_date, order_id), show order revenue and the previous completed order revenue for that. Display customer_id, order_id, order_date, order_revenue, prev_order_revenue.
9. For completed orders ordered by order_date and then by order_id, compute the running sum of order_revenue. Display order_id, order_date, order_revenue, running_revenue.
10. Within each completed order, rank line items by line revenue descending. Return lines with line_rank 1 or 2 only.
11. For each completed order, show order_revenue, the overall average completed-order revenue, and the difference from that average. Display order_id, order_date, order_revenue, avg_completed_order_revenue, diff_from_avg.